# Amy Ashurst Gooch

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## Education



**Northwestern University, Evanston, Illinois**
Doctor of Philosophy in Computer Science, June 2006.
Title: *Preserving Salience By Maintaining Perceptual Differences for Image Creation and Manipulation*.
Advisor: Professor Jack Tumblin.



**University of Utah, Salt Lake City, Utah**
Master of Science in Computer Science, December 1998.
Thesis title: *Interactive Non-photorealistic Technical Illustration*.
Advisors: Professor Peter Shirley and Professor Elaine Cohen.



**University of Utah, Salt Lake City, Utah**
Bachelor of Science in Computer Engineering, June 1996.

## Employment



**ViSUS, Salt Lake City, Utah** **August 2013–present**
*Research Scientist and Chief Operations Officer*

The ViSUS software framework was designed with the primary philosophy that the visualization of massive data need not be tied to specialized hardware or infrastructure. In other words, a visualization environment for large data can be designed to be lightweight, highly scalable and run on a variety of platforms or hardware. As ViSUS Chief Operations Officer, I lead the commercialization and development of ViSUS products and coordinate projects with all ViSUS employees, as well as with our research institution and national laboratory collaborators. I have more than 22 years in image processing and user interaction design/evaluation and 8 years developing commercial software for web/mobile devices; I also use my experience with 8 apps on Apple's App Store with 2.8 million users to push ViSUS products into usable products. Between selling applications and course work form the NSF Innovation Corp on Lean Start-ups and entrepreneurial, I learned much about customers, their needs and how to match the two into solutions. I am the PI for an NSF SBIR Phase II, a DOE SBIR Phase II and a DOE STTR/SBIR Phase I. My primary employment (100% of work week) is at the small business.



**University of Utah, Salt Lake City, Utah** **August 2013–present**
*Research Scientist, Scientific Computing and Imaging Institute*

Working with large code base; change programmer built User Interfaces into user-friendly configurations. We use JUCE to create device agnostic apps on platforms such as iOS/MacOSX/Windows. I collaborate with scientist in other disciplines, such as the doctors at University of Utah's Center for Alzheimer's Care, Imaging and Research (CACIR) to create a prototype of an iOS app for memory care and 3D microscopy software for the Neuroscience Department. Helped write grants for NIH STTR Phase II, NSF SBIR Phase I, NSF AIR-TT, DOE STTR/SBIR. Awarded grant for Entrepreneurial Lead in NSF Innovation Corps (I-Corp) Program, April 2015. Submitted and support Gigapixel David app on Apple App Store.

**Insatiable Genius LLC.** Las Vegas, NV **July 2011– present**
Chief Operations Officer

Insatiable Genius LLC. specializes development of iPhone apps and has found great success in their creativity enhancing products. Insatiable Genius works by the motto imagine, create, and share. They subscribe to Pablo Picasso's philosophy that "everything you can imagine is real." Their personal curiosity and the urge to share their discoveries with the world result in technically superior, innovatively unique and artistically creative software.

GraffitiTagger iPhone/iPad app: I worked on the UI design and layout; as well as a third of the programming; created tutorial video

ToonPAINT HD: port ToonPAINT to iPad; New UI layout

ComicTycoon: create version for iPhone and a version for iPad; Coached a team of programmers and artists through app idea to product;

PencilFX: worked with team of 11 people through creating new engine and an iOS universal app.

**University of Victoria, Victoria, British Columbia, Canada** **September 2006 – Present**
*Assistant Professor, Department of Computer Science*

Image creation is an attempt to record a scene for presentation to a viewer. However, there is a precarious mapping between how the scene would be perceived and how the images are perceived. The human visual system does not have an absolute sense of color or intensity. Instead it is more accurate to say that the human visual system interprets images based on the relative difference between image regions. The goal of my research is to create more salient images by preserving perceptual differences between image regions during image creation and manipulation.

**Northwestern University, Evanston, IL** **August 2004–September 2006**
*Ph.D. Student, Department of Computer Science*

**Northwestern University, Evanston, IL** **August 2003–September 2006**
*Lecturer, School of Continuing Studies, Department of Computer Science, and School of Communication*

Create several courses to be taught to high school students (summer), undergraduates, graduates students, as well as professionals seeking Masters degrees. Please see Teaching Summary for a detailed list of courses.

**Northwestern University, Evanston, IL** **February 2004–June 2004**
*Consulting, Department of Psychology*

Creating software and visual stimuli, as well as handling hardware such as an eye-tracker, galvanic skin response, and heart rate monitors, for perception and action research with Professor Peter Vishton.

**University of Utah, Salt Lake City, Utah** **November 1998–July 2003**
*Computer Graphics Researcher, VisSim Research Group*

As a Computer Graphics Researcher I worked on the design and implementation of two immersive environments. The first is a one-of-a-kind treadmill surrounded by 3 screens (210 degrees field of view). The second immersive environment consists of a nVision head-mounted display device with InterSense trackers (10'x20'). For both of these projects I developed the virtual reality software using OpenGl, WorldToolKit, and C++ on LINUX, SGI, SUN, and HP machines. I have also modeled and textured several virtual worlds using Alias|Wavefront's Maya modeling software. I worked with several cognitive psychologists to create perception experiments to evaluate the user's sense of scale, distance, and slope within these immersive environments. I was responsible for researching, evaluating, and purchasing both software and hardware. I created videos from live feed and from computer displays, as well as maintain web-based documentation for the research group's software projects. I also supervised four undergraduate research interns for per year during Summer 2000, 2001, and 2002, 2003.

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**University of Utah, Salt Lake City, Utah** **Summer 1995–November 1998**
*Research Assistant, Alpha1 Research Group*

As a research assistant during my undergraduate and graduate career, I conducted research and assisted in the development and maintenance of the Alpha_1 NURBS-based geometric modeling and manufacturing system. I created a renderer for generating non-photorealistic technical illustration of NURBS-based models using Tcl/Tk and C++ on SGI, SUN, and HP machines. I also created and maintained web-based documentation for Alpha_1 research system, and collaborated with Brown University on the Sketch/JOT project.

## Additional Start-Up Companies

**Relevant Games LLC.**, Founder and Board Member
Sept. 2009–2013

Relevant Games LLC. specializes in the creation of casual games with socially relevant themes. Titles include "Save the Frogs" and "Spice Islands." Relevant Games are served using social media applications (Facebook) and on mobile devices (Microsoft Mobile, Android, and iPhone).

## Publications

### Books

[1] *Non-Photorealistic Rendering*, Bruce Gooch and Amy Gooch. AK Peters Ltd, Natick, MA. (254 pages hardback)., July 2001. ISBN: 978-1568811338.

[2] *Non-Photorealistic Rendering*, Stuart Green, David Salesin, Simon Schofield, Aaron Hertzmann, Peter Litwinowicz, Amy Gooch, Cassidy Curtis, and Bruce Gooch. ACM SIGGRAPH 99 Course # 17; published in paperback conference course notes book available for purchase and available on ACM Digital Library, 1999. A. Gooch contribution includes 8-1 – 8-17 and slides: 8-1 – 8-24; 10-1 – 10-12 and slides: 10-1 – 10-22.

[3] *Computer-Generated Medical, Technical, and Scientific Illustration*, Mario Coast Sousa, David Ebert, Amy Gooch, and Don Stredney. ACM SIGGRAPH 2005 Course #31; published in paperback conference course notes book available for purchase and available on ACM Digital Library, 2005. A. Gooch contribution includes pages 1 – 114.

### Journal Articles

PDFs available at http://www.csc.uvic.ca/ agooch/publication.html

[4] Visual cues for perceiving distances from objects to surfaces. Helen Hu, Amy A. Gooch, Sarah Creem-Regehr, and William B. Thompson. *Journal of Presence: Teleoperators and Virtual Environments*, 11(6):652–664, December 2002.

[5] Perceiving Virtual Geographical Slant: Action Influences Perception. Sarah Creem-Regehr, Amy A. Gooch, Cynthia Sahm, and William Thompson. *Journal of Experimental Psychology: Human Perception and Performance*, 30(5):811–821, 2004.

[6] Human facial illustrations: Creation and Psychophysical Evaluation. Bruce Gooch, Erik Reinhard, and Amy Gooch. *ACM Transactions on Graphics*, 23(1):27–44, 2004. Google Scholar: cited by 109.

[7] Does the Quality of Computer Graphics Matter When Judging Distances in Visually Immersive Environments? William B. Thompson, Peter Willemsen, Amy A. Gooch, Sarah Creem-Regehr, Jack M. Loomis, and Andrew C. Beall. *Journal of Presence: Teleoperators and Virtual Environments*, 13(5):560–571, October 2004. Google Scholar: cited by 120.

[8] The Influence of Restricted Viewing Conditions and Egocentric Distance Perception: Implications for Real and Virtual environments. Sarah H. Creem-Regehr, Peter Willemsn, Amy A. Gooch, and William B. Thompson. *Perception*, 34:191–204, 2005.

[9] Color2Gray: Salience-Preserving Color Removal. Amy A. Gooch, Sven C. Olsen, Jack Tumblin, and Bruce Gooch. *ACM Transactions on Graphics*, 24(3):634–639, 2005. Google Scholar: Cited by 81.

[10] Semanticons: Visual Metaphors as File Icons. Vidya Setlur, Conrad Albrecht-Buehler, Amy Gooch, Sam Rossoff, and Bruce Gooch. *Computer Graphics Forum*, 24(3):647–656, 2005.

[11] Visualizing pentimenti: revealing the hidden history of paintings. Amy A. Gooch and Jack Tumblin. *Journal of Mathematics and the Arts*, 1(2):133–142, 2007.

[12] Effects of stereo viewing conditions on distance perception in virtual environments. P. Willemsen, A.A. Gooch, W.B. Thompson, and S.H. Creem-Regehr. *Presence: Teleoperators and Virtual Environments*, 17(1):91–101, 2008.

[13] Automatically mimicking unique hand-drawn pencil lines. Zainab AlMeraj, Brian Wyvill, Tobias Isenberg, Amy A. Gooch, and Richard Guy. *Computers & Graphics*, 33(4):496–508, 2009. Erdös number of 2 for authoring with Richard Guy.

[14] Abstraction of man-made shapes. Ravish Mehra, Qingnan Zhou, **Jeremy Long**, Alla Sheffer, Amy Gooch, and Niloy J. Mitra. *ACM Transactions on Graphics*, pages 137:1–137:10, 2009. first appeared in ACM SIGGRAPH Asia 2009 Conference Proceedings.

[15] Resolution independent npr-style 3d line textures. Kristin Potter, Amy Gooch, Bruce Gooch, Peter Willemsen, Joe Kniss, Richard Riesenfeld, and Peter Shirley. *Computer Graphics Forum*, 28(1):56–66, 2009.

[16] Paint and click: Unified interactions for image boundaries. B. Summa, A. A. Gooch, G. Scorzelli, and V. Pascucci. *Computer Graphics Forum*, 34(2):385–393, 2015.

**Refereed Conference Articles**

PDFs available at http://www.csc.uvic.ca/ agooch/publication.html

[17] A Non-photorealistic Lighting Model for Automatic Technical Illustration. Amy Gooch, Bruce Gooch, Peter Shirley, and Elaine Cohen. In *Proceedings of the 25th annual conference on Computer graphics and interactive techniques*, SIGGRAPH '98, pages 447–452, New York, NY, USA, 1998. ACM. ACM Siggraph '98 Conference Proceedings, Google Scholar: cited by 316.

[18] Interactive technical illustration. Bruce Gooch, Peter-Pike J. Sloan, Amy Gooch, Peter Shirley, and Richard Riesenfeld. In *I3D '99: Proceedings of the 1999 symposium on Interactive 3D graphics*, pages 31–38, New York, NY, USA, 1999. ACM Press. Google Scholar: cited by 212.

[19] Visual Cues for Imminent Object Contact in Realistic Virtual Environments. Helen H. Hu, Amy A. Gooch, William B. Thompson, Brian E. Smits, John J. Rieser, and Peter Shirley. In *Proceedings of the conference on Visualization '00*, VIS '00, pages 179–185, Los Alamitos, CA, USA, 2000. IEEE Computer Society Press.

[20] The lit sphere: a model for capturing npr shading from art. Peter-Pike J. Sloan, William Martin, Amy Gooch, and Bruce Gooch. In *Graphics interface 2001*, pages 143–150, 2001.

[21] Perceiving virtual geographical slant: action influences perception. Sarah H. Creem-Regehr, Amy A. Gooch, and William B. Thompson. *Journal of Vision (Abstract)*, 2(7), 2002.

[22] Evaluating Space Perception in NPR Immersive Environments. Amy Gooch and Peter Willemsen. In *Proceedings of the 2nd international symposium on Non-photorealistic animation and rendering*, NPAR '02, pages 105–110, New York, NY, USA, 2002. ACM.

[23] Perceived Distance in Virtual Environments: An Experimental Comparison between Real, Image-Based, and Traditional Virtual Environments. Peter Willemsen and Amy Gooch. In *Poster Paper, Proceedings of IEEE Virtual Reality*, pages 275–276, 2002.

[24] The effects of restricted viewing conditions on egocentric distance judgments. Sarah Creem-Regehr, Peter Willemsen, Amy Gooch, and William Thompson. *Journal of Vision (Abstract)*, 3(9), 2003.

[25] Compression of distance judgments when viewing virtual environments using a head-mounted display. William Thompson, Amy Gooch, Peter Willemsen, Sarah Creem-Regehr, Jack Loomis, and Andrew Beall. *Journal of Vision (Abstract)*, 3(9), 2003.

[26] Enhancing perceived depth in images via artistic matting. Amy Gooch and Bruce Gooch. In *Poster paper in Proceedings of the 1st Symposium on Applied perception in graphics and visualization*, APGV '04, page 168, New York, NY, USA, 2004. ACM.

[27] Enhancing perceived depth in images via artistic matting. Amy Gooch and Bruce Gooch. *Eurographics Workshop on Computational Aesthetics*, pages 83–90, May 2005.

[28] Illustrative Scientific Visualization Framework. Mario Costa Sousa, Amy Gooch, and Bruce Gooch. *Eurographics Workshop on Computational Aesthetics*, pages 57–66, May 2005.

[29] Interweaving gaming into core cs curriculum. Yolanda Rankin, Bruce Gooch, and Amy Gooch. *Second International Conference on Game Development in Computer Science Education*, pages 22–28, February 2007.

[30] Aesthetics of graph visualization. Jody Ryall, Chris Bennett, Leo Spalteholz, and Amy Gooch. *Eurographics International Conference on Computational Aesthetics*, pages 57–64, June 20-22 2007.

[31] Individual realities: Individual and group aesthetic judgments in immersive virtual reality. David Sprague and Amy Gooch. *Eurographics International Conference on Computational Aesthetics*, pages 143–149, June 20-22 2007.

[32] Mimicking hand-drawn pencil lines. Z. AlMeraj, B. Wyvill, T. Isenberg, A.A. Gooch, and Richard Guy. In *CAe '08: In Proceedings of the International Symposium on Computational Aesthetics in Graphics, Visualization, and Imaging. (CAe 2008, June 180, 2008, Lisbon, Portugal)*, pages 73–80, Aire-la-Ville, Switzerland, 2008. Eurographics Association.

[33] Impact of game design on students' interest in cs. Yolanda Rankin, Amy Gooch, and Bruce Gooch. *Third International Conference on Game Development in Computer Science Education*, pages 31–35, March 2008.

[34] Addressing industry issues in a multi-disciplinary course on game design. **Anthony Estey**, Amy Gooch, and Bruce Gooch. In *Proceedings of the 4th International Conference on Foundations of Digital Games*, FDG '09, pages 71–78, New York, NY, USA, 2009. ACM.

[35] One-click white balance using human skin reflectance. **Jeremy Long** and Amy A. Gooch. In *GI '09: Proceedings of Graphics Interface 2009*, pages 55–62, Toronto, Ont., Canada, Canada, 2009. Canadian Information Processing Society.

[36] Investigating the application of a studio-based learning pedagogoical approach into a course on game design. **Anthony Estey**, **Jeremy Long**, Bruce Gooch, and Amy A. Gooch. In *Proceedings of the Fifth International Conference on the Foundations of Digital Games*, FDG '10, pages 64–71, New York, NY, USA, 2010. ACM.

[37] Towards mapping the field of non-photorealistic rendering. Amy A. Gooch. In *Proceedings of the 8th International Symposium on Non-Photorealistic Animation and Rendering*, NPAR '10, pages 159–164, New York, NY, USA, 2010. ACM.

[38] Viewing progress in non-photorealistic rendering through heinlein's lens. Amy A. Gooch, **Jeremy Long**, **Anthony Estey**, **Ji Li**, and Bruce Gooch. In *Proceedings of the 8th International Symposium on Non-Photorealistic Animation and Rendering*, NPAR '10, pages 165–171, New York, NY, USA, 2010. ACM.

[39] Catalyst: seeing through the eyes of a cat. **J. Long**, **A. Estey**, D. Bartle, S. Olsen, and A. A. Gooch. In *Proceedings of the Fifth International Conference on the Foundations of Digital Games*, FDG '10, pages 116–123, New York, NY, USA, 2010. ACM.

[40] Recovering color from black and white photographs. Sven Olsen, Rachel Gold, Amy Gooch, and Bruce Gooch. In *Computational Photography (ICCP), 2010 IEEE International Conference on*, pages 1–8, March 2010.

[41] Bee prepared: Simulating bee vision in a game environment. **Jeremy Long** and Amy Gooch. In *16th annual IEEE Computer Games Conference*, 2011. Received the Best paper award.

[42] Double meandering algorithm: From drawing game to automated animation. **Shelley Gao**, Lucy Pullen, and Amy Gooch. In *Accepted for CAe '11: In Proceedings of the International Symposium on Computational Aesthetics in Graphics, Visualization, and Imaging. (CAe 2011, August 5-7, Vancouver, BC)*, Vancouver, BC, 2011. Eurographics Association.

[43] Sound anchoring: Content-based exploration of music collections with anchored self-organized maps. L. Collares, T. F. Tavares, J. Feliciano, S. Gao, G. Tzanetakis, and A. Gooch. In *Proc. Sound and Music Computing (SMC)*, 2013.

[44] Morphable guidelines for the human head. Shelley Gao, Christopher Werner, and Amy A. Gooch. In *Proceedings of the Symposium on Computational Aesthetics*, CAE '13, pages 21–28, New York, NY, USA, 2013. ACM.

**Other Articles**

PDFs available at http://www.csc.uvic.ca/ agooch/publication.html

[45] Implicit color field. **Li Ji**, **Nick Cohea**, and Amy Gooch. *Poster in Proceeding of Non-Photorealistic Animation and Rendering, NPAR 2011*, 2012. Received the Best Poster award.

## Patents

Process and Systems for Relighting Images using Reflectance Matching. **Jeremy Long** and Amy Ashurst Gooch. Provisional patent, June 2009, Docket #9007P. App. No. 61359986. Patent Pending.

## Funding

### Fellowship:

Helen and Robert J. Piros Fellowship 2004-2005

### Grants:

| Funding obtained by ViSUS and Dr. Amy Gooch. | | | | |
|---|---|---|---|---|
| Agency | Project Title | Dates | Type | Amount |
| NSF Innovation I-Corps | Commercialization of Multiplatform Gigapixel Imaging and Annotation Software, Award Abstract #1542353, PI Attila Gyulassy, Amy Gooch Co-PI | 05/15/2015 - 10/31/2015 | I-Corp | $50,000 |
| NSF SBIR Phase I | "Immediate Delivery of Massive Aerial Imagery to Farmers and Crop Consultants", Grant 1549187 | 01/01/2016 to 01/31/2017 | SBIR/STTR | $150,000 |
| NSF SBIR Phase II | "Immediate Delivery of Massive Aerial Imagery to Farmers and Crop Consultants", Grant 1738448 | 09/01/2017 to 08/31/2020. | SBIR/STTR | $746,107 |
| NSF SBIR Phase II CAP Supplement | "Immediate Delivery of Massive Aerial Imagery to Farmers and Crop Consultants", Grant 1738448 | 2019 | SBIR/STTR | $10,000 |
| DOE SBIR/STTR Phase I | "Ease of use and deployment for a fast, scalable data movement infrastructure", Grant DE-SC0017152 | 02/21/2017 through 10/20/2018. | SBIR/STTR | $224,927.00 |
| DOE SBIR/STTR Phase II | "Ease of use and deployment for a fast, scalable data movement infrastructure", Grant DE-SC0017152 | 05/21/2018 - 05/20/2020 | SBIR/STTR | $1,499,218 |
| DOE SBIR/STTR Phase I | "VisStore: Seamless Acquisition, Storage, and Distribution of Massive Imagery blending Cloud, Local and HPC storage.", Grant DE-SC0020549 | 02/18/2020 - 02/17/2021 | SBIR/STTR | $206,499 |

| Research funding obtained in support of Dr. Amy Gooch's research. | | | | |
|---|---|---|---|---|
| NSERC Discovery Grant | Preserving Salience By Maintaining Perceptual Differences for Image Creation and Manipulation | 2012-17 | Cash | $17,000 |
| MITACS and ConversionWorks Collaboration | | April 2011 – | Cash | $15,000 |
| NSERC Discovery Grant | Preserving Salience By Maintaining Perceptual Differences for Image Creation and Manipulation | 2011 | Cash | $18,000 |
| MITACS and Intel Cluster Collaboration (joint B. Wyvill and B. Gooch) | | Jan 2011 – | Cash | $116,666 |
| (joint with B. Wyvill and B. Gooch) | Advanced Graphics Hardware Research | 2010 | Cash | $35,000 |
| NSERC Discovery Grant | Preserving Salience By Maintaining Perceptual Differences for Image Creation and Manipulation | 2010-11 | Cash | $18,000 |
| Networks of Centres of Excellence of Canada (part of $8.4 million/yr, shared with 50 NI and 50 CNI) | GRAND: Graphics, Animation, and New Media | 2010 – | Cash | $11,600/year |
| (Joint B. Gooch) | Semantic Icons | Feb. 2009 | Cash | $8,500 |
| (Joint B. Wyvill and B. Gooch) | Advanced Graphics Hardware Research | April 2009 – April 2010 | Cash | $35,000 |
| NSERC Discovery Grant | Preserving Salience By Maintaining Perceptual Differences for Image Creation and Manipulation | 2009-10 | Cash | $18,000 |
| (Joint B. Wyvill, B. Gooch, N. Horspool) | Advanced Graphics Hardware Research | April 2008 – April 2009 | Cash | $35,000 |

| (Continued) Research funding obtained in support of Dr. Amy Gooch's research. | | | | |
|---|---|---|---|---|
| Agency | Project Title | Dates | Type | Amount |
| NSERC USRA | Art and CS Collabroation (Shelley Gao) | May. 2009 – Aug. 2009 | Cash | $4,500 |
| NSERC Discovery Grant | Preserving Salience By Maintaining Perceptual Differences for Image Creation and Manipulation | 2008-09 | Cash | $18,000 |
| | Save the Frogs: Entertaining Educational Games for Zoo Exhibits (Donna Kaminskyj, Elyse Regan, Kyliah Clarkson) http://uvicugrads.blogspot.com/ | Sept. 2008 – Sept. 2009 | Cash | $9,500 |
| NSERC USRA | Games for CS (Donna Kaminskyj, Elyse Regan) | May. 2008 – Aug. 2008 | Cash | $9,000 |
| | Colour Science for Computational Photography | May. 2008 – May 2013 | Cash | $82,000 |
| BCKDF CFI Matching | Colour Science for Computational Photography | May. 2008 – May 2013 | Cash | $82,000 |
| CRA-W CDMP Ugrad Research | "Pointer for Freshman: Teaching Computer Science Concepts through Educational Games". (Donna Kaminskyj, Elyse Regan) http://uvicugrads.blogspot.com/ | May. 2008 – Sept. 2009 | Cash | $1000 Honorium + 2 funded students ($7000 for 16 weeks) |
| NSERC Discovery Grant | Preserving Salience By Maintaining Perceptual Differences for Image Creation and Manipulation | 2007-08 | Cash | $18,000 |
| | Google Earth Competition: UVic Model (Brittany Sims, Susan Perkins) | May. 2007 – Aug. 2007 | Cash | $9,000 for two students |

| (Continued) Research funding obtained in support of Dr. Amy Gooch's research. | | | | |
|---|---|---|---|---|
| Agency | Project Title | Dates | Type | Amount |
| CRA-W CDMP Ugrad Research | "Pointer for Freshman: Teaching Computer Science Concepts through Educational Games". (Susan Perkins) | May. 2007 – Sept. 2007 | Cash | $4,000 for students |
| CRA-W CDMP Ugrad Research | "Pointer for Freshman: Teaching Computer Science Concepts through Educational Games". (Susan Perkins) | May. 2007 – Sept. 2007 | Cash | $1,000 honorium |
| NSERC Discovery Grant | Preserving Salience By Maintaining Perceptual Differences for Image Creation and Manipulation | 2006-2007 | Cash | $18,000 |
| | UVic Professional Development Faculty Award | May. 2007 – May. 2008 | Cash | $2,743 |
| | UVIC IRG Top-Up Award | May. 2007 – May. 2008 | Cash | $675 |
| | UVIC Startup Grant | 2006-07 | Cash | $50,000 |
| | Autodesk Research Donation: 9 seats of Maya Unlimited Rendering and Animation, Cleaner, Combustion, and 3D Studio Max | Sept. 2006 | In-Kind | $180,000 |
| | Games on Multiple Hardware Platforms | 2005–2006 | Cash | $80,000 |
| | Autodesk Research Donation: 9 seats of Maya Unlimited Rendering and Animation, Cleaner, Combustion, and 3D Studio Max | Sept. 2001 – Present | In-Kind | $160,000 |

*Note: Canadian funding is unique in two key ways 1) Grants do not incur overhead. 2) Industry donations receive 2-to-1 matching from federal and provincial sources.*

## Teaching Summary



**Department of Computer Science, University of Victoria, Victoria, B.C. Canada**

Spring 2012, CSC 305: Introduction to Computer Graphics

Spring 2012, CSC 299: Directed Independent Study (Colton Hash)

Fall 2011, CSC 205: 2D Computer Graphics and Image Processing

Fall 2011, CSC 490: Directed Studies (Li ”Ray” Rui)

Summer 2011, CSC 486: Generating Animations: Software & Scripting

Summer 2011, CSC 305: Introduction to Computer Graphics

Spring 2011, CSC 299: Directed Studies (Connor Sauve)

Spring 2011, CSC 499: Honors Project (Tim Song)

Spring 2011, CSC 499: Honors Project (Tyler Smith)

Spring 2011, CSC 499: Honors Project (Kristian Hein)

Spring 2011, CSC 486D: Special Topics: Computational Aesthetics

Fall 2010, CSC 499: Directed Studies (Carl Sverre)

Fall 2010, CSC 205: 2D Computer Graphics and Image Processing

Summer 2010, CSC 499: (Directed Studies): Andrew Martin: Games and Metaballs

Summer 2010, CSC 305: Introduction to Computer Graphics

Spring 2010, CSC 205: 2D Computer Graphics and Image Processing

Spring 2010, CSC 595: Graduate Seminar

Fall 2009, CSC 499: (Directed Studies), Shelley Gao: Animation and Scripting

Fall 2009, CSC 499: (Directed Studies), Matthew Paxman: Animation and Scripting

Fall 2009, CSC 499: (Directed Studies), Anna Macaranas: Animation and Scripting

Fall 2008, CSC 485: Generating Animations: Software & Scripting

Fall 2008, CSC 212: Practical Computer Science

Summer 2008, CSC 499: (Directed Studies), Tom Spreen ”CubeRoids” `http://spreenblog.blogspot.com/`

Summer 2008, CSC 305: Introduction to Computer Graphics

Spring 2008, CSC 305: Introduction to Computer Graphics

Fall 2007, CSC 591: (Directed Studies), Sven Olsen, Multispectral Imagery for Computational Photography

Fall 2007, CSC 591: (Directed Studies), Sam Rossoff, Multispectral Imagery for Computational Photography

Fall 2007, CSC 591: (Directed Studies), Jeremy Long, Multispectral Imagery for Computational Photography

Fall 2007, CSC 212: Practical Computer Science

Spring 2007, CSC 485: Special Topics: Computational Aesthetics



**School of Communication, Northwestern University, Evanston, Illinois**
Fall 2005, co-teaching CAT 380/480 *Animate Art III*. Introduce 3D geometry, shading, lighting, and Maya Embedded Language (MEL) as part of a 4-quarter curriculum in computer-based art and entertainment systems. Fall quarter focuses on 3D modeling and interactive simulation. Student's work is critiqued both as works of art and as engineering. The interaction between art practice and art theory will be addressed through discussions, critiques and readings, examining issues in the interpretation, understanding, and production of art and visual culture.



**Department of Computer Science, Northwestern University, Evanston, Illinois**
Spring 2006, co-teaching CS 395/495 *Game Authoring Tools*. A software-engineering class disguised as a course which focuses on game production and development for multiple hardware platforms. Students will build their own 2D and 3D games on top of available game engines.

Spring 2005, CS 395/495 *Introduction to Computer Generated Animation*. An introduction to computer generated animation thru the Alias' Maya software package. Students are introduced to modeling, shading, lighting, and character set-up and animation.



**School of Continuing Studies, Northwestern University, Evanston, Illinois**
Fall 2003, Fall 2004, instructor for SCS COMP SCI 351, *Introduction to Computer Graphics*, a graduate level course examining the mathematical software and hardware requirements for computer graphics systems. Special attention is paid to data structures and programming languages.

Winter 2004, instructor for SCS COMP SCI 395, *Special Topics: Advanced Computer Graphics*, a graduate level seminar course. Students read about, present, and discuss advanced graphics techniques.

Summer 2005, 2004, SCS COMP SCI 395, *Introduction to Computer Generated Animation*. An introduction to computer generated animation thru the Alias' Maya software package. Students are introduced to modeling, shading, lighting, and character set-up and animation.



**University of Utah, Salt Lake City, Utah**
Fall 2002, lecturer in CS 5540, *Human/Computer Interaction*, a undergraduate course on human factors in the context of designing interactive applications.

Spring 2002, guest lecturer in ART 4980, *Senior Studio Seminar*, a undergraduate art course divided into two areas: creative process and professional concerns for the working artist.

Spring 2001, guest lecturer in ART 6810, *MFA Graduate Seminar*, a graduate art course divided into two areas: creative process and professional concerns for the working artist.

Spring 2001, guest lecturer in ART 4980, *Senior Studio Seminar*, a undergraduate art course divided into two areas: creative process and professional concerns for the working artist.

Winter 1997, Teaching Assistant, *Advanced Manufacturing*, Department of Computer Science, University of Utah. Tutored mechanical engineering students in the use of CAD/CAM modeling system. Aided in the creation of models and images.

Summer 1997, Teaching Assistant, *High School Summer Computing Institute*, Department of Computer Science, University of Utah. Taught high school students to use a CAD/CAM modeling system in an intensive six week summer program. Guided and encouraged students in the design of high level geometric models.

Fall 1996, Teaching Assistant, *Introduction to Unix*, Department of Computer Science, University of Utah. Assisting college students on a web-based course, both on-line and face to face.

### Tutorials and Symposia


ACM SIGGRAPH 2005 Course: Computer Generated Medical, Technical and Scientific Illustration.


Disney Feature Animation Inc., June 1999, Symposium: *Non-Photorealistic Rendering*


ACM Siggraph 1999, Course: *Non-Photorealistic Rendering*

## Instructor Development

Attended Learning and Teaching Center one week workshop on Course Redesign (Aug 18th-22nd, 2008) (Certificate Awarded)

Attended Microsoft Faculty Summit 2008; There are several sessions on how to increase enrollment, increase retention, new programs added to Computer Science to attract students.

Attended workshop on Game Curriculum Development (Microsoft Academic Days 2008)

Attended talk by Jamie Cormack (Microsoft) on Assessment and Teaching 2008

Attended Teaching Buffet, Nov 13th, 2007

Attended workshop on Game Curriculum Development (Microsoft Academic Days 2007)

Attended Microsoft Faculty Summit 2007; There are several sessions on how to increase enrollment, increase retention, new programs added to Computer Science to attract students.

## University Service

Co-created new courses approved by faculty for addition in the 2009-2010 calendar year

- CSC 205: 2D Computer Graphics and Image Processing
- CSC 471: Fundamentals of Computer Rendering
- CSC 472: Fundamentals of Computer Modeling
- CSC 473: Fundamentals of Computer Animation
- CSC 486: Special Topics in Computer Graphics

Created new special topics courses: Computational Aesthetics, Generating Animations: Software and Scripting

UVic Computer Science: Mentor for four Freshman (2007-2008)

UVic Computer Science Academic Integrity Committee 2007

Chair the oral exam of Master's candidate Leslie Anderson (Biology) on Wednesday, April 11, 2007

UVic Computer Science Academic Integrity Committee 2008

Chair the oral exam of Master's candidate Leila Scannell (Psychology) on Monday, April 21, 2008.

Panelist Northwestern University 35th Annual Career Day for Girls, February 25, 2006.

Panelist Northwestern University 34rd Annual Career Day for Girls, February 26, 2005.

Panelist Northwestern University 33rd Annual Career Day for Girls, February 21, 2004.

## Community Service: Reviewer

NSF Electronic Proposal (2004)

ACM Journal Transactions on Applied Perception

ACM Journal Transactions on Graphics

ACM Non-Photorealistic Animation and Rendering Conference

ACM SIGGRAPH Conference , ACM SIGGRAPH ASIA Conference

Eurographics Conference, Full Papers, Short Papers

Computers & Graphics: An International Journal

IEEE Visualization

IEEE Computer Graphics and Applications

IEEE Transactions on Visualization and Computer Graphics

Eurographics Symposium on Rendering (EGSR)

Graphics Interface, Graphics and HCI Track

Computer Graphics Forum

MidGraph 2004, 2005

Eurographics Workshop on Computational Aesthetics

Second International Conference on Game Development in Computer Science Education (2007)

Eurographics Workshop on Computational Aesthetics Artist's Submissions

Graphics Hardware

Eurographics Symposium on Rendering

IEEE Transactions on Pattern Analysis and Machine Intelligence

Foundations of Digital Games conference

## Other Community Service

Program chair of Graphics Track for Graphics Interface 2009, co-chair of paper with Dr. Melanie Tory

- creation of program committee
- creation of web page, advertising, CFP
- inviting guest speakers
- appoint Student Volunteer Chair, Publicity Chair
- administer paper reviewing and selection process in a one day session located in Victoria
- serve as liaison between authors of accepted papers and editor for proceedings
- organizing papers into sessions, selecting session chairs, create conference schedule
- select best student paper awards
- oversee technical sessions on-site and make sure paper authors are registered and show up

Program committee member: Second International Conference on Game Development in Computer Science Education 2007

Program committee member: Graphics Interface 2007, 2008

Program committee member: Computational Aesthetics 2007, 2008, 2009, 2010

Program Committee, Schoss Dagstuhl, Int. Conf. and Research Center for Computer Science, Special Meeting on Computational Aesthetics 2008

Program Committee of NPAR 2009, 2010 (International Symposium on Non-Photorealistic Animation and Rendering)

Program Committee for Foundations of Digital Games conference special track on games in computer science education.

Program Committee of the 5th International Symposium on Visual Computing (ISVC09 - http://www.isvc.net)

## Invited Talks

"Expanding Our Vision: Images and Gaming", University of Denver, February 2010

"Job Search Process", ACM CRA-W Grad Co-hort, Seattle, WA March 2008

"Graphics at UVic", Invited talk at University of British Columbia Feb 2008

"Graphics at UVic", Invited talk at University of Utah 2007

"CS1: Where's It Going, and What Should We Be Thinking About?", Microsoft Faculty Summit, July 2007

"Preserving Salience By Maintaining Perceptual Differences for Image Creation and Manipulation". Workshop on CG: Current Trends in Research and Industry, Lahore University, Pakistan, July 9th, 2007

"Revealing Pentimenti: the hidden history of paintings", SPIE Human Vision and Electronic Imaging Conference. Invited talk and invited paper, San Jose CA, Feb. 2007

"Visualizing Pentimenti: Revealing Hidden History in Paintings". Dagstuhl Seminar on Computational Aesthetics in Graphics, Visualization and Imaging. May 2006.

"Color2Gray: Salience-Preserving Color Removal". University of Tubigen. May 2006.

"Preserving Salience By Maintaining Perceptual Differences for Image Creation and Manipulation". Purdue University. April 2006.

"Preserving Salience By Maintaining Perceptual Differences for Image Creation and Manipulation". College of William and Mary. March 2006.

"Preserving Salience By Maintaining Perceptual Differences for Image Creation and Manipulation". University of Victoria. February 2006.

"Color2Gray: Salience-Preserving Color Removal". University of Utah. July 2005.

"Color2Gray: Salience-Preserving Color Removal", ACM SIGGRAPH 2005.

"Color2Gray: Salience-Preserving Color Removal". Computational Aesthetics. May 18-20th, 2005. Girona, Spain. http://www.computational-aesthetics.org/html/links.html

"Illustration: Lighting and Material Properties". ACM SIGGRAPH 2005, part of the SIGGRAPH 2005 Computer Generated Medical, Technical and Scientific Illustration Half-Day Course.

"Perception and Computer Graphics". Washington University, Computer Science Department Colloquium. November 2003.

"Perception and Computer Graphics". University of Iowa, Department of Computer Science Colloquium, February 2003.

"Perception and Computer Graphics". University of Pennsylvania, GRASP lab, Dr. Kostas Daniilidis. January 2003

"Non-Photorealistic Rendering". Disney Feature Animations Disney-Graph, June 1999.

"Using Non-Photorealistic Rendering to Communicate Shape". ACM Siggraph 1999, part of the SIGGRAPH 1999 Non-Photorealistic Rendering Full-day Course.

"A Non-Photorealistic Lighting Model For Automatic Technical Illustration", ACM Siggraph 1998 paper.

## Participant

NSF Innovation Corps (I-Corp) Program, April 2015 GATech Cohort, an 8 week intensive program intended to help bridge the gap between academic artifacts and commercial enterprises

Dagstuhl Seminar on Computational Aesthetics in Graphics, Visualization and Imaging (May 28 - June 6, 2006)

Contributor to the first ACM/Eurographics Workshop on Computational Aesthetics (May 2005)

Co-organizer for MidGraph, the MidWest Computer Graphics Meeting (2003, 2004)

Finance Chair for the first Symposium on Graphics and Perception to be held in conjunction with ACM SIGGRAPH 2004

Contributor to the first ACM Campfire on Graphics and Perception

## Graduate Students

Anthony Estey, MSc Student (2008– July 2010)

Ryan Wallace, MSc Student (2009–2011)

Jeremy Long, Ph.D. student(2007–2012)

Ji Li, Ph.D. Student (2009 – present)

Shelley Gao , MSc Student(2010 – 2014)

Leandro Collares, MSc Student (2011 – 2014) (co-advised with Dr. Yvonne Coady)

## Research Undergraduate Students

Susan Perkins Summer 2007

Britany Sims, Summer 2007

Donna Kaminskyj, Summer 2008–Summer 2009

Eylse Regan, Summer 2008–Summer 2009

Kyliah Clarkson, Summer 2008–Summer 2009

Shelley Gao, Summer 2009

Shelley Gao, Summer 2010

Nick Cohea, Summer 2010

Li (Ray) Rui, Fall 2011

Jessie Su, Fall 2011

Hyerin (Helen) Hong, Fall 2011

Colton Hash, Spring 2012

## Computer Science References

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## Interdisciplinary Research References

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Professor John J. Rieser
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